

Zero-Norm in Support Vector Machine

H.A. LE THI, M.C. NGUYEN

Laboratory of Theoretical and Applied Computer Science
UFR MIM, University of Lorraine, Ile du Saulcy, 57045 Metz, France

1 SVM with l_0 -norm regularization

2 MSVM with l_0 -norm regularization

2.1 MSVM generic model

Let W is a set of Q vectors in R^d , B is a vector in R^Q and ξ is a set of n vectors in R^Q . Let X is a input set which includes n vectors x_i in R^d and Y is a vector in R^n which contains the class labels.

The MSVM training problem (generic model) which is proposed by Guermeur as follows:

$$\min_{w,b,\xi} \frac{1}{2} \|w\|_q^r + C \|\xi\|_p^p, \quad (1)$$

st.

$$\Omega : \begin{cases} \langle K1.w_{y_i} - w_k, x_i \rangle + K1.b_{y_i} - b_k \geq K2 - \xi_{ik}, & \forall i \in \overline{1,n}; \forall k \in \overline{1,Q}; k \neq y_i \\ K3(\xi_{ik} - \xi_{il}) = 0, & \forall i \in \overline{1,n}; \forall k, l \in \overline{1,Q}; k, l \neq y_i \\ (2-p)\xi_{ik} \geq 0, & \forall i \in \overline{1,n}; \forall k \in \overline{1,Q}; k \neq y_i \\ (1-K1) \sum_{k=1}^Q h_k = 0, & \forall k \in \overline{1,Q}. \end{cases}$$

Here, we consider the cases $p=1$. In this problem, $r=2$ in the case $q=2$; and $r=1$ in others. In general, with the set of the values of $K1$, $K2$, $K3$, we have 3 models as follow:

- The model Weston & Watkins: $K1 = 1; K2 = 1; K3 = 0$.
- The model Cramer & Singer : $K1 = 1; K2 = 1; K3 = 1$.
- The model Lee & Lin : $K1 = 0; K2 = 1/(Q-1); K3 = 0$.

2.2 l_0 -norm and l_2l_0 -norm regularization

• l_0 -norm regularization problem

$$\min_{w,b,\xi} \frac{1}{2} \|w\|_0 + C \|\xi\|_1^1, \quad (2)$$

$$s.t. (w, b, \xi) \in \Omega.$$

$$\Leftrightarrow \min_{w,b,\xi} \frac{1}{2} \sum_{i=1}^Q \|w_i\|_0 + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik}, \quad (3)$$

$$s.t. (w, b, \xi) \in \Omega.$$

- l_2l_0 -norm regularization problem

$$\min_{w,b,\xi} \beta \|w\|_2^2 + \frac{1}{2} \|w\|_0 + C \|\xi\|_1, \quad (4)$$

$$s.t. (w, b, \xi) \in \Omega.$$

$$\Leftrightarrow \min_{w,b,\xi} \beta \sum_{i=1}^Q \|w_i\|_2^2 + \frac{1}{2} \sum_{k=1}^Q \|w_k\|_0 + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik}, \quad (5)$$

$$s.t. (w, b, \xi) \in \Omega.$$

In the section below, we apply 7 methods to approximate the l_0 term of the problems (3) and (5); and the DCAs to solve these resulting optimization problems.

2.3 DCA1: DC formulations and DCA for concave approximation

First, let us describe some notations and background materials that will be used in the sequel. The base of the natural logarithm will be denoted by ε and for a vector $x \in R^n$, ε^{-x} will denote a vector in R^n with components ε^{-x_i} , $i = 1, \dots, n$. For $w \in R^n$, w^* is the step vector defined by

$$w_i^* = 1 \text{ if } w_i \neq 0; 0 \text{ otherwise.}$$

Hence, the zero-norm $\|w\|_0$ can be written as $e^T w^*$. For $x \in R^n$ and a given $\alpha > 0$, let η be the function defined by

$$\eta(x) = \begin{cases} 1 - \varepsilon^{-\alpha x} & \text{if } x \geq 0, \\ 1 - \varepsilon^{\alpha x} & \text{if } x \leq 0. \end{cases}$$

The step vector w^* , for all $w \in R^n$ can be approximated by $w_i^* \simeq \eta(w_i)$. and then, an approximation of the zero-norm $\|w\|_0$ would be $\|w\|_0 \simeq \sum_{i=1}^n \eta(w_i)$. With this approximation, the l_0 -norm in (3) and (5) is approximated by

$$\|w_i\|_0 \simeq \sum_{j=1}^d \eta(w_{ij}).$$

The resulting optimization problem of (3) takes the form:

$$\min_{(w,b,\xi) \in \Omega} \left\{ C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \sum_{i=1}^Q \sum_{j=1}^d \eta(w_{ij}) \right\}, \quad (6)$$

and the corresponding form of (5):

$$\min_{(w,b,\xi) \in \Omega} \left\{ C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \beta \sum_{k=1}^Q \|w_k\|_2^2 + \sum_{i=1}^Q \sum_{j=1}^d \eta(w_{ij}) \right\}. \quad (7)$$

- DC formulations

We are finding now a DC decomposition for the objective function of (6). First, we express η as a DC function

$$\eta(x) = g_1(x) - h_1(x),$$

where

$$g_1(x) = \begin{cases} \alpha x & \text{if } x \geq 0 \\ -\alpha x & \text{if } x \leq 0 \end{cases}; \quad h_1(x) = \begin{cases} \alpha x - 1 + \varepsilon^{-\alpha x} & \text{if } x \geq 0 \\ -\alpha x - 1 + \varepsilon^{\alpha x} & \text{if } x \leq 0. \end{cases}$$

The objective function of (6) can be now written as:

$$C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \sum_{i=1}^Q \sum_{j=1}^d g_1(w_{ij}) - \sum_{i=1}^Q \sum_{j=1}^d h_1(w_{ij}).$$

Set

$$G_1(w, b, \xi) = C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \sum_{i=1}^Q \sum_{j=1}^d g_1(w_{ij}) \quad (8)$$

and

$$H_1(w, b, \xi) = \sum_{i=1}^Q \sum_{j=1}^d h_1(w_{ij}). \quad (9)$$

So, the problem (6) is equivalent to:

$$\min \{G_1(w, b, \xi) - H_1(w, b, \xi) : (w, b, \xi) \in \Omega\}. \quad (10)$$

Obviously, g_1 and h_1 are convex functions, and so are G_1 and H_1 . Therefore (10) is a DC program. Note that since $g_1(x) = \max(\alpha x, -\alpha x)$, the function g_1 is polyhedral convex and so is G_1 . Therefore (10) is a polyhedral DC program.

We are now in the position of finding a DC formulation for the problem (7). Let G_2 be a function defined by

$$G_2(w, b, \xi) = G_1(w, b, \xi) + \beta \sum_{i=1}^Q \|w_i\|_2^2. \quad (11)$$

Clearly, G_2 is convex. The optimization problem (7) can be written as

$$\min \{G_2(w, b, \xi) - H_1(w, b, \xi) : (w, b, \xi) \in \Omega\}, \quad (12)$$

and (12) is a DC program.

• DCA for (10) and (12)

We are now developing a version of DCA to solve the DC program (10). According to the generic DCA scheme, applying DCA on (10) amounts to computing, at each iteration k , a subgradient $Y^k(\bar{w}, \bar{b}, \bar{\xi})$ of $H_1(X^k(w^k, b^k, \xi^k))$ and then, solve the convex program:

$$\min \{G_1(w, b, \xi) - \langle Y^k, X \rangle : X(w, b, \xi) \in \Omega\}. \quad (13)$$

So, $Y^k = (\bar{w}^k, 0, 0)$ with, $i = 1, \dots, Q; j = 1, \dots, d$

$$\bar{w}_{ij}^k = \begin{cases} \alpha(1 - \varepsilon^{-\alpha w_{ij}^k}) & \text{if } w_{ij} \geq 0 \\ -\alpha(1 - \varepsilon^{\alpha w_{ij}^k}) & \text{if } w_{ij} < 0, \end{cases} \quad (14)$$

and the convex program (13) in this situation is

$$\min \left\{ G_1(X) - \sum_{i=1}^Q \sum_{j=1}^d \bar{w}_{ij}^k w_{ij}, (w, b, \xi) \in \Omega \right\} \quad (15)$$

$$\Leftrightarrow \min \left\{ C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \sum_{i=1}^Q \sum_{j=1}^d \max(\alpha w_{ij}, -\alpha w_{ij}) - \sum_{i=1}^Q \sum_{j=1}^d \bar{w}_{ij}^k w_{ij}, (w, b, \xi) \in \Omega \right\} \quad (16)$$

$$\Leftrightarrow \min_{w, b, \xi, t} \left\{ C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \sum_{i=1}^Q \sum_{j=1}^d t_{ij} - \sum_{i=1}^Q \sum_{j=1}^d \bar{w}_{ij}^k w_{ij} \right\} \quad (17)$$

$$s.t. \quad (w, b, \xi) \in \Omega, t_{ij} \geq \alpha w_{ij}, t_{ij} \geq -\alpha w_{ij}, i = 1, \dots, Q; j = 1, \dots, d.$$

Finally, the DC Algorithm for the problem (10) is described as follows:

The Algorithm 1: l_0 -DCA1

Initializations:

Let $\epsilon > 0$ be given, $X^0(w^0, \xi^0, b^0)$ be an initial point. Select α and set $k = 0$;

Repeat:

1. Calculate $\bar{w}^k$ from w^k via (14)
2. Calculate X^{k+1} from Y^k : solve the convex program (17)
3. $k = k + 1$.

Until $\|X^{k+1} - X^k\| \leq \epsilon \|X^k\|$.

Remark 2.1 Since G_1 is polyhedral convex, l_0 -DCA1 has a finite convergence.

Similarly, the DC algorithm for the problem (12) can be described as l_0 -DCA1 with β is given, but at each iteration k , instead of solving (17), we solve the quadratic convex optimization problem

$$\min_{w, b, \xi, t} C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \beta \sum_{i=1}^Q \|w_i\|_2^2 + \sum_{i=1}^Q \sum_{j=1}^d t_{ij} - \sum_{i=1}^Q \sum_{j=1}^d \bar{w}_{ij}^k w_{ij} \quad (18)$$

$$s.t. \quad (w, b, \xi) \in \Omega, t_{ij} \geq \alpha w_{ij}, t_{ij} \geq -\alpha w_{ij}, i = 1, \dots, Q; j = 1, \dots, d.$$

Here, the parameters for turning are C, α, β .

• **Implementations**

The Algorithm 2: l_2 - l_0 -DCA1**Initializations:**

Let $\epsilon > 0$ be given, $X^0(w^0, \xi^0, b^0)$ be an initial point. Select α, β and set $k = 0$;

Repeat:

1. Calculate $\bar{w}^k$ from w^k via (14)
2. Calculate X^{k+1} from Y^k : solve the convex program (18)
3. $k = k + 1$.

Until $\|X^{k+1} - X^k\| \leq \epsilon \|X^k\|$.

We choose the value of $\alpha = 0.9$ and the choice of β is 0.1. To select features, we compute the feature ranking score $c_j, j = 1, \dots, d$ for each feature. Let $w_i, i = 1, \dots, Q$ be the weight vector of the i th hyperplane and $w_{ij}, i = 1, \dots, Q, j = 1, \dots, d$ be the weight value of hyperplane i th associated with the feature j th. We first normalize the weight vectors

$$w_i = \frac{w_i}{\|w_i\|}, i = 1, \dots, Q.$$

The ranking score c_j is calculated by $c_j = \frac{1}{Q} \sum_{i=1}^Q w_{ij}^2$, and we remove the features j for which c_j is smaller than a given threshold ρ . After removing features, for computing the accuracy of classification, we apply again l_2 -MSVM on the new training datasets and calculate the classification's accuracy on the new test sets.

CONCAVE PENALTY: L2L0																		
Thres.:		0.001				0.005				0.01				0.015				
Data	Feat.	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	CPU	
LUN	56	7	56.25	56.25	7	56.25	56.25	7	56.25	56.25	7	56.25	56.25	7	56.25	56.25	0.12	
MOV	90	50	47.62	69.52	33	7.62	75.324	29	6.67	73.33	22	6.67	75.24	22	6.67	75.24	193.7	
CNA	856	82	87.07	88.62	54	81.38	83.79	29	68.28	69.13	20	58.28	58.97	20	58.28	58.97	16.62	
HIL	100	6	100	99.51	6	100	99.5	6	100	99.5	5	100	99.8	5	100	99.8	5.85	
SPA	57	26	87.65	86.52	20	89.43	87.61	16	89.43	87.57	12	87.48	83.78	12	87.48	83.78	0.44	
ADV	1558	19	76.13	94.03	15	76.39	93.55	12	76.13	93.11	10	77.53	93.24	10	77.53	93.24	4.22	
ADN	60	13	73.8	82.75	4	80.96	80.87	4	80.96	80.87	4	80.96	80.87	4	80.96	80.87	1.99	

Table 1. number of selected features (SF), accuracy (%) and CPU of L_2 - L_0 -DCA1

CONCAVE PENALTY: L0																		
Thres.:		0.001				0.005				0.01				0.015				
Data	Feat.	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	CPU	
LUN	56	8	43.75	56.25	8	43.75	56.25	8	43.75	56.25	8	43.75	56.25	8	43.75	56.25	0.18	
MOV	90	48	40	74.29	35	20	76.19	27	18.09	75.24	21	14.29	72.38	169.58				
CNA	856	81	88.1	88.28	48	76.55	80.89	28	63.1	70.35	24	63.1	67.76	17.77				
HIL	100	4	97.03	100.00	4	97.03	100.00	4	97.03	100.00	4	97.03	100.00	5.85				
SPA	57	27	90.22	87.70	15	89.09	89.57	8	83.48	47.00	7	83.48	47.00	0.47				
ADV	1558	15	69.55	90.52	15	69.55	90.52	15	69.55	90.52	15	69.55	90.52	4.14				

Table 2. number of selected features (SF), accuracy (%) and CPU of L_0 -DCA1

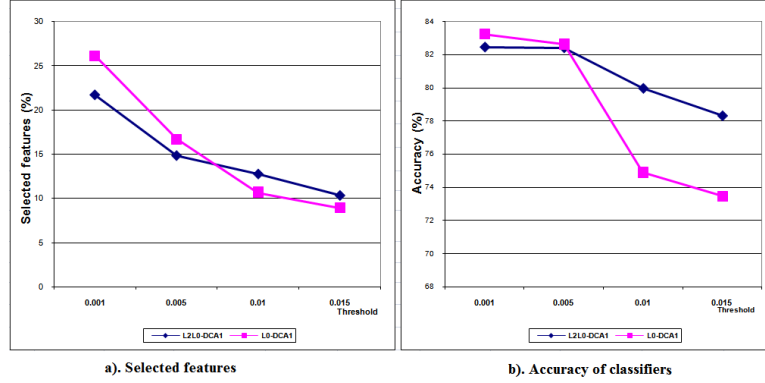


Fig. 1. Feature selection performance of the methods on each dataset.

2.4 DCA2: approximation of the l_0 -norm by SCAD penalty function

In this section, to approximate the l_0 -norm in (3) and (5), we consider the SCAD penalty $p_\lambda(|t|)$ proposed by Fan and Li and define the function f_λ which coincides with p_λ for all $t \geq 0$, namely:

$$f_\lambda(t) = \begin{cases} \lambda t & \text{if } 0 \leq t \leq \lambda, \\ -\frac{t^2 - 2\alpha\lambda t + \lambda^2}{2(\alpha-1)} & \text{if } \lambda < t \leq \alpha\lambda, \\ \frac{(\alpha+1)\lambda^2}{2} & \text{if } t > \alpha\lambda, \\ f_\lambda(-t) & \text{if } t < 0, \end{cases}$$

where $\lambda > 0$ and $\alpha > 2$. The problem (3) is equivalent to

$$\min_{w, b, \xi} \left\{ C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \sum_{i=1}^Q \sum_{j=1}^d f_\lambda(w_{ij}); (w, b, \xi) \in \Delta \right\}, \quad (19)$$

and the problem (5) is equivalent to

$$\min_{w, b, \xi} \left\{ C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \sum_{i=1}^Q \sum_{j=1}^d f_\lambda(w_{ij}); (w, b, \xi) \in \Delta \right\}. \quad (20)$$

• DC decomposition and ADC for 19

First, let us show a DC decomposition of $f_\lambda(t)$. Let g_2 and h_2 be the functions given by

$$g_2(t) = \begin{cases} \lambda t & \text{if } t \geq 0 \\ -\lambda t & \text{if } t < 0, \end{cases}$$

and

$$h_2(t) = \begin{cases} 0 & \text{if } 0 \leq t \leq \lambda, \\ \frac{1}{2(\alpha-1)}(t - \lambda)^2 & \text{if } \lambda < t \leq \alpha\lambda, \\ \lambda t - \frac{(\alpha+1)\lambda^2}{2} & \text{if } t > \alpha\lambda, \\ h_2(-t) & \text{if } t < 0. \end{cases}$$

Clearly g_2 and h_2 are two convex functions and $f_\lambda(t) = g_2(t) - h_2(t)$. Then we have the following DC decomposition of (19):

$$\min\{G_3(w, b, \xi) - H_3(w, b, \xi) : (w, b, \xi) \in \Omega\}, \quad (21)$$

where

$$G_3(w, b, \xi) = C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \sum_{i=1}^Q \sum_{j=1}^d g_2(w_{ij}) \quad (22)$$

and

$$H_3(w, b, \xi) = \sum_{i=1}^Q \sum_{j=1}^d h_2(w_{ij}) \quad (23)$$

are convex.

Applying DCA scheme on (21) amounts to computing, at each iteration k , a subgradient v^k of $H_3(w^k)$ and then, solve the convex program:

$$\min\{G_3(w, b, \xi) - \langle v^k, w \rangle : (w, b, \xi) \in \Omega\} \quad (24)$$

Clearly, v^k can be chosen as

$$v_{ij}^k = \begin{cases} 0 & \text{if } -\lambda \leq w_{ij}^k \leq \lambda, \\ (\alpha - 1)^{-1}(w_{ij}^k - \lambda) & \text{if } \lambda < w_{ij}^k \leq \alpha\lambda, \\ (\alpha - 1)^{-1}(w_{ij}^k + \lambda) & \text{if } -\alpha\lambda \leq w_{ij}^k < -\lambda, i = 1, \dots, Q; j = 1, \dots, d. \\ \lambda & \text{if } w_{ij}^k > \alpha\lambda, \\ -\lambda & \text{if } w_{ij}^k < -\alpha\lambda, \end{cases} \quad (25)$$

and the convex program (24) is equivalent to the following linear program

$$\min_{w, b, \xi, \eta} \left\{ C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \sum_{i=1}^Q \sum_{j=1}^d \eta_{ij} - \sum_{i=1}^Q \sum_{j=1}^d v_{ij}^k w_{ij} \right\} \quad (26)$$

$$s.t. (w, b, \xi) \in \Omega, \quad \lambda w_{ij} \leq \eta_{ij}, \quad -\lambda w_{ij} \leq \eta_{ij}, i = 1, \dots, Q; j = 1, \dots, d.$$

Finally, the DCA applied to (21) can be described as follows:

The Algorithm 3: l_0 -DCA2

Initializations: Let $\epsilon > 0$ be sufficiently small. Choose an initial point $u^0(\xi^0, w^0, b^0)$. Select C, α, λ and set $k = 0$;

Repeat:

- Calculate $v^k \in \partial H_1(w^k)$ via (25);
- Calculate $u^{k+1}(\xi^{k+1}, w^{k+1}, b^{k+1})$ by solving the linear program (26);
- $k = k + 1$.

Until $\|u^{k+1} - u^k\| \leq \epsilon \|u^k\|$.

Remark 2.2 Since G_3 is polyhedral convex, l_0 -DCA2 has a finite convergence.

• DC decomposition and ADC for 20

In the same way, the problem (20) can be reformulated as:

$$\min\{G_4(w, b, \xi) - H_3(\xi, w, b) : (w, b, \xi) \in \Omega\}, \quad (27)$$

where $G_4(w, b, \xi) = G_3(w, b, \xi) + \beta \sum_{i=1}^Q \|w_i\|_2^2$. G_4 is a convex function and the DCA for (27) can be described as l_0 -DCA2 with β is given, but at each iteration k , instead of solving (26), we solve the quadratic convex optimization problem

$$\min_{w, b, \xi, \eta} C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \beta \sum_{i=1}^Q \|w_i\|_2^2 + \sum_{i=1}^Q \sum_{j=1}^d \eta_{ij} - \sum_{i=1}^Q \sum_{j=1}^d v_{ij}^k w_{ij} \quad (28)$$

s.t. $(w, b, \xi) \in \Omega$, $\lambda w_{ij} \leq \eta_{ij}$, $-\lambda w_{ij} \leq \eta_{ij}$, $i = 1, \dots, Q$; $j = 1, \dots, d$.

The Algorithm 4: l_2 - l_0 -DCA2

Initializations: Let $\epsilon > 0$ be sufficiently small. Choose an initial point $u^0(\xi^0, w^0, b^0)$. Select $C, \alpha, \lambda, \beta$ and set $k = 0$;

Repeat:

- Calculate $v^k \in \partial H_1(w^k)$ via (25);
- Calculate $u^{k+1}(\xi^{k+1}, w^{k+1}, b^{k+1})$ by solving the quadratic convex program (28);
- $k = k + 1$.

Until $\|u^{k+1} - u^k\| \leq \epsilon \|u^k\|$.

• **Implementations**

SCAD PENALTY: L2L0														
Thres.:		0.001			0.005			0.01			0.015			
Data	Feat.	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	CPU
LUN	56	6		56.25	5		56.25	5		56.25	4		56.25	0.16
MOV	90	57		74.29	49		76.19	36		75.24	25		63.81	151.23
CNA	856	56		88.79	42		87.93	34		86.21	28		84.66	50.55
HIL	100	22		100.00	11		100.0	10		100.0	9		97.7	13.79
SPA	57	29		88.39	14		90.22	10		89.52	7		88.22	45.12
ADV	1558	8		95.31	6		95.00	9		95.00	9		95.00	5.29
ADN	60	21		85.44	4		80.96	4		80.96	4		80.96	62.67

Table 3. number of selected features (SF), accuracy (%) and CPU of L_2 - L_0 -DCA2

SCAD PENALTY: L0														
Thres.:		0.001			0.005			0.01			0.015			
Data	Feat.	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	CPU
LUN	56	31		50.00	11		56.25	7		62.5	7		62.5	0.12
MOV	90	50		73.33	26		71.43	19		74.29	17		73.33	69.55
CNA	856	57	92.59	87.07	42	85.52	87.76	35	80.86	86.21	30	67.07	86.72	25.51
HIL	100	14	53.14	100.00	12	48.68	100.00	10	71.29	100.0	8	48.68	100.0	10.474
SPA	57	16	90.48	87.61	14	88.91	87.17	13	88.91	85.48	12	86.39	86.00	1.55
ADV	1558	75	91.27	93.77	38	89.6	93.37	19	87.23	91.31	13	91.36	91.36	3.28
ADN	60	31	79.5	87.04	12	80.96	85.25	7	80.96	86.24	4	80.96	80.44	1.62

Table 4. number of selected features (SF), accuracy (%) and CPU of L_0 -DCA2

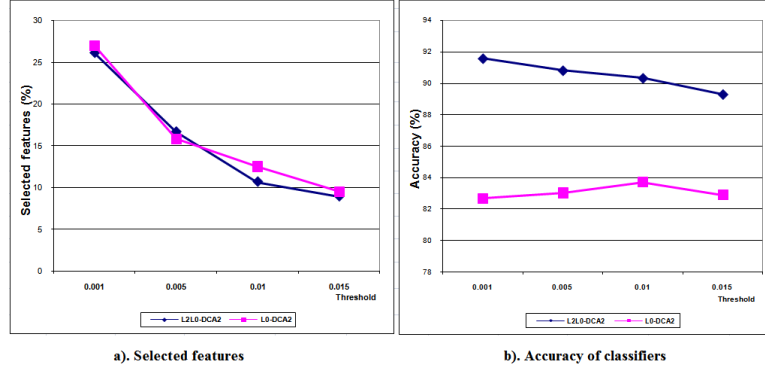


Fig. 2. Feature selection performance of the methods on each dataset.

2.5 DCA3: new approximated function

The $\|\cdot\|_0$ is approximated by: $\|w\|_0 \simeq \sum_{i=1}^Q \sum_{j=1}^d t(w_{ij})$ where

$$t(x) = \min\{1, \alpha|x|\} = 1 + \alpha|x| - \max\{1, \alpha|x|\}. \quad (29)$$

So the objective function of (3) is equivalent to

$$\min_{w, b, \xi} \left\{ \sum_{i=1}^Q \sum_{j=1}^d t(w_{ij}) + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} \right\}. \quad (30)$$

We first present the DC decomposition of $t(x)$: $t(x) = g(x) - h(x)$ where $g(x) = 1 + \alpha|x|$ and $h(x) = \max\{1, \alpha|x|\}$. Clearly $g(x)$ and $h(x)$ are convex functions and (3) is equivalent to

$$\min_{w, b, \xi} \left\{ \left[\sum_{i=1}^Q \sum_{j=1}^d g(w_{ij}) \right] + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} - \left[\sum_{i=1}^Q \sum_{j=1}^d h(w_{ij}) \right] \right\}, X(w, b, \xi) \in \Delta, \quad (31)$$

and in detail, (3) can be written as

$$\min_{w, b, \xi} \left\{ \left[\sum_{i=1}^Q \sum_{j=1}^d (1 + \alpha|w_{ij}|) \right] + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} - \left[\sum_{i=1}^Q \sum_{j=1}^d \max\{1, \alpha|w_{ij}|\} \right] \right\}, X(w, b, \xi) \in \Delta, \quad (32)$$

We set:

$$G(X) = \sum_{i=1}^Q \sum_{j=1}^d g(w_{ij}) + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} = \sum_{i=1}^Q \sum_{j=1}^d (1 + \alpha|w_{ij}|) + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik}, \quad (33)$$

and

$$H(X) = \sum_{i=1}^Q \sum_{j=1}^d h(w_{ij}) = \sum_{i=1}^Q \sum_{j=1}^d \max\{1, \alpha|w_{ij}|\}, \quad (34)$$

so the problem (3) is equivalent to

$$\min_{w,b,\xi} \{G(X) - H(X), X(w, b, \xi) \in \Delta\}, \quad (35)$$

where $G(X)$ and $H(X)$ are two convex functions.

Applying DCA on (35) amounts to computing, at each iteration k , a subgradient Y^k of $H(X^k)$ and solve the convex program: $\min\{G(X) - \langle X^k, X \rangle : X(w, b, \xi) \in \Delta\}$

Calculation of $\nabla H(X)$

$$\frac{\partial H}{\partial \xi_{ij}} = 0, \frac{\partial H}{\partial b_j} = 0; \quad i = 1, \dots, n; \quad j = 1, \dots, Q,$$

$$\frac{\partial H}{\partial w_{ij}} = \begin{cases} 0 & \text{if } -1/\alpha \leq w_{ij} \leq 1/\alpha \\ \alpha & \text{if } w_{ij} > 1/\alpha \\ -\alpha & \text{if } w_{ij} < -1/\alpha \end{cases}$$

So, $Y^k(\bar{w}^k, \bar{b}^k, \bar{\xi}^k) \in \nabla H(X^k)$ is calculated by

$$\bar{\xi}_{ij}^k = 0; \bar{b}_i^k = 0. \quad (36)$$

$$\bar{w}_{ij}^k = \begin{cases} 0 & \text{if } -1/\alpha \leq w_{ij}^k \leq 1/\alpha \\ \alpha & \text{if } w_{ij}^k > 1/\alpha \\ -\alpha & \text{if } w_{ij}^k < -1/\alpha \end{cases} \quad (37)$$

Calculation of $X^{k+1} \in \partial G^*(Y^k)$

$$X^{k+1} = \arg \min_{w,b,\xi} \left\{ G(X) - \sum_{i=1}^Q \sum_{j=1}^d \bar{w}_{ij}^k w_{ij}, (w, b, \xi) \in \Delta \right\} \quad (38)$$

$$= \arg \min_{w,b,\xi} \left\{ \frac{1}{2} \sum_{i=1}^Q \sum_{j=1}^d \alpha |w_{ij}| + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} - \sum_{i=1}^Q \sum_{j=1}^d \bar{w}_{ij}^k w_{ij}, (w, b, \xi) \in \Delta \right\} \quad (39)$$

$$= \arg \min_{w,b,\xi,t} \left\{ \frac{1}{2} \sum_{i=1}^Q \sum_{j=1}^d t_{ij} + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} - \sum_{i=1}^Q \sum_{j=1}^d \bar{w}_{ij}^k w_{ij} \right\} \quad (40)$$

$$s.t. (w, b, \xi) \in \Delta, t_{ij} \geq \alpha w_{ij}, t_{ij} \geq -\alpha w_{ij}, i = 1, \dots, Q; j = 1, \dots, d.$$

DCA3 for the problem (3) is described as follows:

Initializations: Let $\epsilon > 0$ be given, $X^0(w^0, \xi^0, b^0)$ be an initial point. Select α and set $k = 0$;

Repeat:

1. Calculate $Y^k(\bar{w}^k, \bar{\xi}^k, \bar{b}^k)$ from X^k via (36) and (37)
2. Calculate X^{k+1} from Y^k via (40)
3. $k = k + 1$.

Until $\|X^{k+1} - X^k\| \leq \epsilon \|X^k\|$. (STOP condition)

Here, the parameters for turning are C .

• DCA3 for the problem (5)

We set

$$\begin{aligned} G_2(X) &= \sum_{i=1}^Q \sum_{j=1}^d g(w_{ij}) + \beta \sum_{i=1}^Q \|w_i\|_2^2 + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} \\ &= \sum_{i=1}^Q \sum_{j=1}^d (1 + \alpha |w_{ij}|) + \beta \sum_{i=1}^Q \|w_i\|_2^2 + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik}, \end{aligned} \quad (41)$$

so, the problem (5) is equivalent to

$$\min_{w, b, \xi} \{G_2(X) - H(X), X(w, b, \xi) \in \Delta\}, \quad (42)$$

and $G_2(X)$, $H(X)$ are two convex functions. The sequence X^{k+1} is calculated by solving the convex optimization problem:

$$\min_{w, b, \xi, t} \left\{ \frac{1}{2} \sum_{i=1}^Q \sum_{j=1}^d t_{ij} + \beta \sum_{i=1}^Q \|w_i\|_2^2 + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} - \sum_{i=1}^Q \sum_{j=1}^d \bar{w}_{ij}^k w_{ij} \right\} \quad (43)$$

$$s.t. (w, b, \xi) \in \Delta, t_{ij} \geq \alpha w_{ij}, t_{ij} \geq -\alpha w_{ij}, i = 1, \dots, Q; j = 1, \dots, d.$$

DCA3 for the problem (5) is described as follows:

Initializations: Let $\epsilon > 0$ be given, $X^0(w^0, \xi^0, b^0)$ be an initial point. Select α and set $k = 0$;

Repeat:

1. Calculate $Y^k(\bar{w}^k, \bar{\xi}^k, \bar{b}^k)$ from X^k via (36) and (37)
2. Calculate X^{k+1} from Y^k : solving problem (43)
3. $k = k + 1$.

Until $\|X^{k+1} - X^k\| \leq \epsilon \|X^k\|$. (STOP condition)

Here, the parameters for turning are C, β .

• Implementations

2.6 DCA4: the exact penalty technique

We replace $\|w\|_0 = \sum_{i=1}^Q \sum_{j=1}^d u_{ij} < e, u >$ with $u \in R^{Q \times d}$ and $u_{ij} = 1$ if $w_{ij} \neq 0$, $u_{ij} = 0$ otherwise. We also use a exact penalty technique for the problems (3) and (5) and these problems can reformulate as follow :

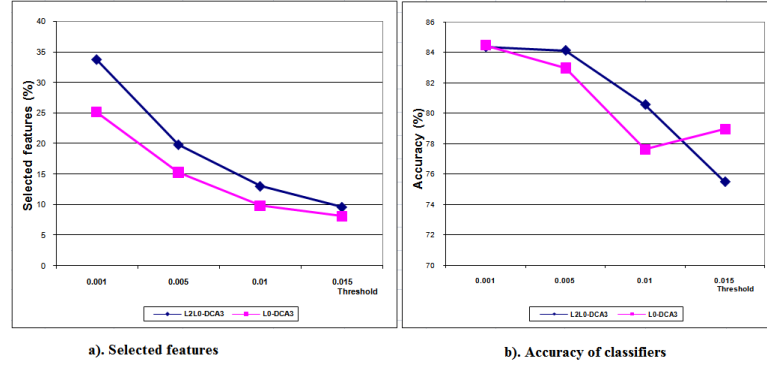
$$(3) \Leftrightarrow \min_{w, b, \xi} \frac{1}{2} \sum_{i=1}^Q \sum_{j=1}^d u_{ij} + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + tp(u) \quad (44)$$

$$s.t. (w, b, \xi) \in \Delta, a_{ij} u_{ij} \leq w_{ij} \leq b_{ij} u_{ij}; 0 \leq u_{ij} \leq 1; i = 1, \dots, Q; j = 1, \dots, d,$$

SCAD PENALTY: L2L0														
Thres.:		0.001			0.005			0.01			0.015			
Data	Feat.	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	CPU
LUN	56	4		56.25	4		56.25	3		56.25	3		56.25	0.072
MOV	90	86		75.24	65		77.14	36		63.81	23		63.81	59.07
CNA	856	89	92.59	91.38	48	85.69	87.93	34	80.86	84.83	26	64.66	83.10	20.15
HIL	100	18	82.51	100.0	12	48.68	100.0	11	48.68	100.0	10	48.68	100.0	4.46
SPA	57	30	89.39	87.83	18	88.52	87.87	13	83.61	85.17	10	83.61	53.48	21.59
ADV	1558	69	91.97	93.73	31	89.47	93.90	22	88.56	93.24	13	87.63	91.53	2.48
ADN	60	29	80.77	86.33	5	78.61	86.05	4	66.16	80.87	3	62.54	80.44	12.21

Table 5. number of selected features (SF), accuracy (%) and CPU of L_2 - L_0 -DCA3

SCAD PENALTY: L0														
Thres.:		0.001			0.005			0.01			0.015			
Data	Feat.	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	CPU
LUN	56	9		62.50	7		56.25	4		50.00	4			0.095
MOV	90	32		71.43	24		70.48	19		73.33	16		60.95	27.37
CNA	856	53	89.48	88.44	42	86.9	87.76	37	77.24	86.55	32	59.14	87.24	4.01
HIL	100	11	50	100.0	8	48.68	100.0	7	48.68	100.0	6	48.68	99.5	3.64
SPA	57	28	90.47	87.96	15	88.78	86.17	11	86.48	53.65	9	83.57	53.35	0.44
ADV	1558	49	92.98	94.91	30	92.37	95.22	25	92.58	95.22	21	90.92	92.45	1.48
ADN	60	33	80.73	86.19	16	76.11	85.12	5	63.95	84.97	3	62.54	80.44	1.14

Table 6. number of selected features (SF), accuracy (%) and CPU of L_0 -DCA3**Fig. 3.** Feature selection performance of the methods on each dataset.

and

$$(5) \Leftrightarrow \min_{w, b, \xi} \frac{1}{2} \sum_{i=1}^Q \sum_{j=1}^d u_{ij} + \beta \sum_{i=1}^Q \|w_i\|_2^2 + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + tp(u) \quad (45)$$

s.t. $(w, b, \xi) \in \Delta$, $a_{ij}u_{ij} \leq w_{ij} \leq b_{ij}u_{ij}$; $0 \leq u_{ij} \leq 1$; $i = 1, \dots, Q$; $j = 1, \dots, d$,

where $a_{ij} = \varepsilon > 0$, $b_{ij} \geq \max_{ij}(w_{ij})$ and $p(u) = \sum_{i=1}^Q \sum_{j=1}^d u_{ij}(1 - u_{ij})$.

So, we have in detail:

$$(3) \Leftrightarrow \min_{w,b,\xi} [C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik}] - [t \sum_{i=1}^Q \sum_{j=1}^d u_{ij}^2 - (t+1) \sum_{i=1}^Q \sum_{j=1}^d u_{ij}], \quad (46)$$

$$s.t. (w, b, \xi) \in \Delta, \quad a_{ij}u_{ij} \leq w_{ij} \leq b_{ij}u_{ij}; \quad 0 \leq u_{ij} \leq 1; \quad i = 1, \dots, Q; \quad j = 1, \dots, d,$$

and

$$(5) \Leftrightarrow \min_{w,b,\xi} [C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \beta \sum_{i=1}^Q \|w_i\|_2^2] - [t \sum_{i=1}^Q \sum_{j=1}^d u_{ij}^2 - (t+1) \sum_{i=1}^Q \sum_{j=1}^d u_{ij}], \quad (47)$$

$$s.t. (w, b, \xi) \in \Delta, \quad a_{ij}u_{ij} \leq w_{ij} \leq b_{ij}u_{ij}; \quad 0 \leq u_{ij} \leq 1; \quad i = 1, \dots, Q; \quad j = 1, \dots, d,$$

We first describe the DC decompositions for these problems. We set:

$$G_1(X) = [C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik}]$$

$$H_1(X) = [t \sum_{i=1}^Q \sum_{j=1}^d u_{ij}^2 - (t+1) \sum_{i=1}^Q \sum_{j=1}^d u_{ij}],$$

and

$$G_2(X) = [C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \beta \sum_{i=1}^Q \|w_i\|_2^2].$$

Clearly $G_1(X), H_1(X), G_2(X)$ are convex functions and:

$$(3) \Leftrightarrow \min_{w,b,\xi} G_1(X) - H_1(X) \quad (48)$$

$$s.t. (w, b, \xi) \in \Delta, \quad a_{ij}u_{ij} \leq w_{ij} \leq b_{ij}u_{ij}; \quad 0 \leq u_{ij} \leq 1; \quad i = 1, \dots, Q; \quad j = 1, \dots, d,$$

and

$$(5) \Leftrightarrow \min_{w,b,\xi} G_2(X) - H_1(X) \quad (49)$$

$$s.t. (w, b, \xi) \in \Delta, \quad a_{ij}u_{ij} \leq w_{ij} \leq b_{ij}u_{ij}; \quad 0 \leq u_{ij} \leq 1; \quad i = 1, \dots, Q; \quad j = 1, \dots, d,$$

• **Solve the problem (3)**

DCA apply to (48) can be described as follow: at each iteration k compute the two sequences $Y^k(\bar{w}^k, \bar{b}^k, \bar{\xi}^k, \bar{u}^k)$ and $X^{k+1}(w^{k+1}, b^{k+1}, \xi^{k+1}, u^{k+1})$ until the convergence of X^k as follow:

- **Calculate Y^k from X^k :**

$$\begin{cases} \bar{w}_{ij}^k = \bar{b}_{ij}^k = \bar{\xi}_{ij}^k = 0, \\ \bar{u}_{ij}^k = 2tu_{ij}^k - (t+1). \end{cases} \quad (50)$$

- **Calculate** X^{k+1} **from** Y^k : solve the LP

$$X^{k+1} = \arg \min_{w,b,\xi,u} \left\{ C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} - \sum_{i=1}^Q \sum_{j=1}^d (2t\bar{u}_{ij}^k - (t+1))u_{ij} \right\} \quad (51)$$

$s.t. (w, b, \xi) \in \Delta, a_{ij}u_{ij} \leq w_{ij} \leq b_{ij}u_{ij}; 0 \leq u_{ij} \leq 1; i = 1, \dots, Q; j = 1, \dots, d,$

Here, the parameters for turning are C, a_{ij}, b_{ij}, t .

• **Solve the problem (P2)**

DCA apply to (49) can be described as follow: at each iteration k compute the two sequences $Y^k(\bar{w}^k, \bar{b}^k, \bar{\xi}^k, \bar{u}^k)$ and $X^{k+1}(w^{k+1}, b^{k+1}, \xi^{k+1}, u^{k+1})$ until the convergence of X^k as follow:

- **Calculate** Y^k **from** X^k : use the formula (50),
- **Calculate** X^{k+1} **from** Y^k via:

$$X^{k+1} = \arg \min_{w,b,\xi,u} \left\{ C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} + \beta \sum_{i=1}^Q \|w_i\|_2^2 - \sum_{i=1}^Q \sum_{j=1}^d (2t\bar{u}_{ij}^k - (t+1))u_{ij} \right\} \quad (52)$$

$s.t. (w, b, \xi) \in \Delta, a_{ij}u_{ij} \leq w_{ij} \leq b_{ij}u_{ij}; 0 \leq u_{ij} \leq 1; i = 1, \dots, Q; j = 1, \dots, d,$

Here, the parameters for turning are $C, \beta, a_{ij}, b_{ij}, t$.

• **Implementation**

[1]. The parameter t is chosen by a line search process:

t belong to $\{20, 100, 200, 300, \dots, 1500\}$.

[2]. For the parameters a_{ij}, b_{ij} : all a_{ij} are equal to a ; all b_{ij} are equal to b .

- $b = \max_{ij} w_{ij}, i = 1, \dots, Q; j = 1, \dots, d.$

- $a = -b.$

The choice of b is depended on the type of initialization.

- Random initialization: b is set to a sufficient large value (e.g. $b = 10.000$).

- Initialize by an other method (DCA3, L_1 -MSVM...): $b = \max_{ij} \{w_{ij}^0\}.$

[3]. The appropriate value of C is chosen by a fivefold cross-validation.

[4]. Feature ranking: the values of w are normalized by columns and ranking score for the feature i th is calculated by the average of the squares of w_{ij} on the column i th of w .

[5]. Initial points:

- Random initialization: the values of u belong to $[0, 1]$ or $u_{ij} = 0, \forall i, j.$

- Initialize by other method:

. Execute the initial method to obtain the solution $(w^*, b^*, \xi^*).$

. The values of u are initialized by: $u_{ij} = 0$ if $|w_{ij}^*| < \varepsilon; k$ otherwise, with

2.7 DCA5: approximation of the l_0 -norm by Log function

Here

Thres.:		0.001			0.005			0.01			0.015			
Data	Feat.	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	CPU
LUN	56	4		56.25	4		56.25	3		56.25	3		56.25	0.072
MOV	90	86		75.24	65		77.14	36		63.81	23		63.81	59.07
CNA	856	89	92.59	91.38	48	85.69	87.93	34	80.86	84.83	26	64.66	83.10	20.15
HIL	100	18	82.51	100.0	12	48.68	100.0	11	48.68	100.0	10	48.68	100.0	4.46
SPA	57	30	89.39	87.83	18	88.52	87.87	13	83.61	85.17	10	83.61	53.48	21.59
ADV	1558	69	91.97	93.73	31	89.47	93.90	22	88.56	93.24	13	87.63	91.53	2.48
ADN	60	29	80.77	86.33	5	78.61	86.05	4	66.16	80.87	3	62.54	80.44	12.21

Table 7. number of selected features (SF), accuracy (%) and CPU of L_2 - L_0 -DCA4

Thres.:		0.001		0.005		0.01		0.015						
Data	Feat.	SF	ACC1	ACC2	SF	ACC1	ACC2	SF	ACC1	ACC2	CPU			
LUN	56	9		62.50	7		56.25	4		50.00	4	0.095		
MOV	90	32		71.43	24		70.48	19		73.33	16	60.95	27.37	
CNA	856	53	89.48	88.44	42	86.9	87.76	37	77.24	86.55	32	59.14	87.24	4.01
HIL	100	11		50 100.0	8	48.68	100.0	7	48.68	100.0	6	48.68	99.5	3.64
SPA	57	28	90.47	87.96	15	88.78	86.17	11	86.48	53.65	9	83.57	53.35	0.44
ADV	1558	49	92.98	94.91	30	92.37	95.22	25	92.58	95.22	21	90.92	92.45	1.48
ADN	60	33	80.73	86.19	16	76.11	85.12	5	63.95	84.97	3	62.54	80.44	1.14

Table 8. number of selected features (SF), accuracy (%) and CPU of L_0 -DCA4

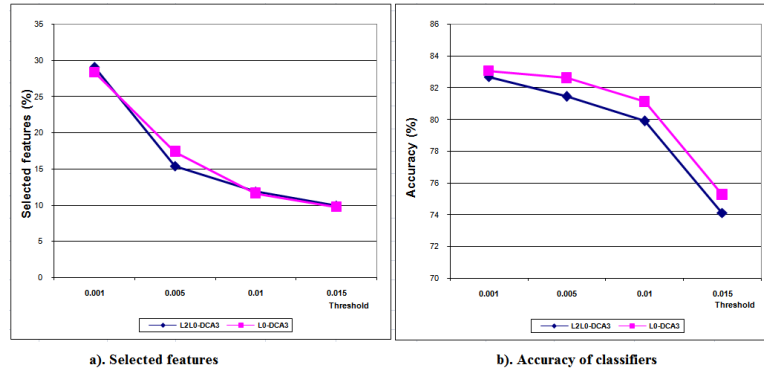


Fig. 4. Feature selection performance of the methods on each dataset.

2.8 CRN1: the convex relaxation

The problem (3) is reformulated as a linear optimization problem with the constraints are relaxed:

$$\min_{w, b, \xi} \frac{1}{2} \sum_{i=1}^Q \sum_{j=1}^d t_{ij} + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} \quad (53)$$

$$s.t. (w, b, \xi, t) \in \Delta; a_{ij}t_{ij} \leq w_{ij} \leq b_{ij}t_{ij}; 0 \leq t_{ij} \leq 1; i = 1, \dots, Q; j = 1, \dots, d,$$

and the problem (5) is a convex quadratic optimization problem with the constraints are relaxed:

$$\min_{w, b, \xi} \frac{1}{2} \sum_{i=1}^Q \sum_{j=1}^d t_{ij} + \beta \sum_{i=1}^Q \|w_i\|_2^2 + C \sum_{i=1}^n \sum_{k \neq y_i} \xi_{ik} \quad (54)$$

$$s.t. (w, b, \xi, t) \in \Delta; a_{ij}t_{ij} \leq w_{ij} \leq b_{ij}t_{ij}; 0 \leq t_{ij} \leq 1; i = 1, \dots, Q; j = 1, \dots, d,$$

Here, the parameters for turning are C, β, a_{ij}, b_{ij} .

References

1. H.A. Le Thi, H.M. Le, V.V. Nguyen, and D.T. Pham, "A DC programming approach for feature selection in support vector machines learning", *ADAC*, 2:259278 (2008).
2. H.A. Le Thi, V.V. Nguyen, and S. Ouchani, "Gene Selection for Cancer Classification Using DCA", —, pp. 62-72 (-).